



FUTURE INTERNS Internship REPORT

Security Operations Center(SOC)
Internship Task: Security Alert
Monitoring & Incident Response
Simulation

Internship Title:

Security Operations Center (SOC) Internship Task: Security Alert Monitoring & Incident Response Simulation

Platform Used:

Splunk Enterprise (Free Trial)

Prepared By: VIJAY S R

Date : 02/11/2025

❖ Objective

The objective of this project is to simulate real-world SOC (Security Operations Center)

Activities such as:

- Monitoring security alerts
- Analyzing potential threats
- Simulating incident response actions

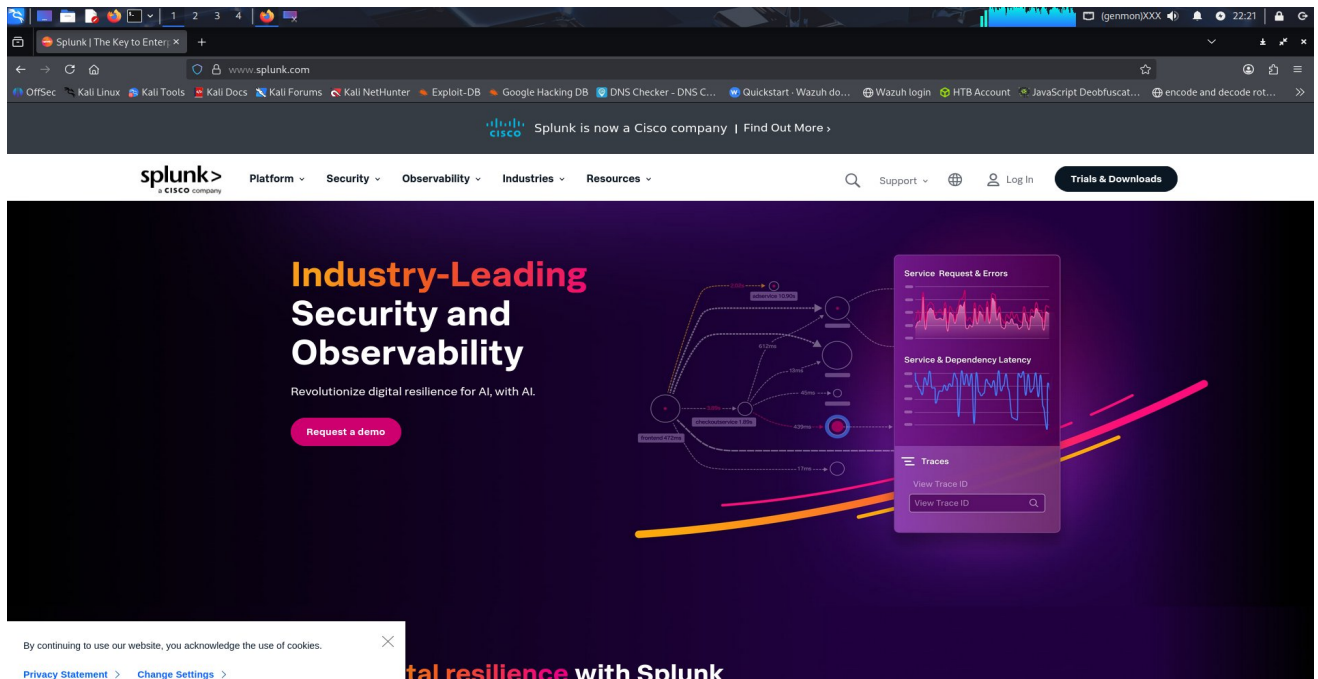
This report demonstrates how Splunk Enterprise can be used for log collection, visualization, alert creation, and incident analysis.

❖ Tools and Setup

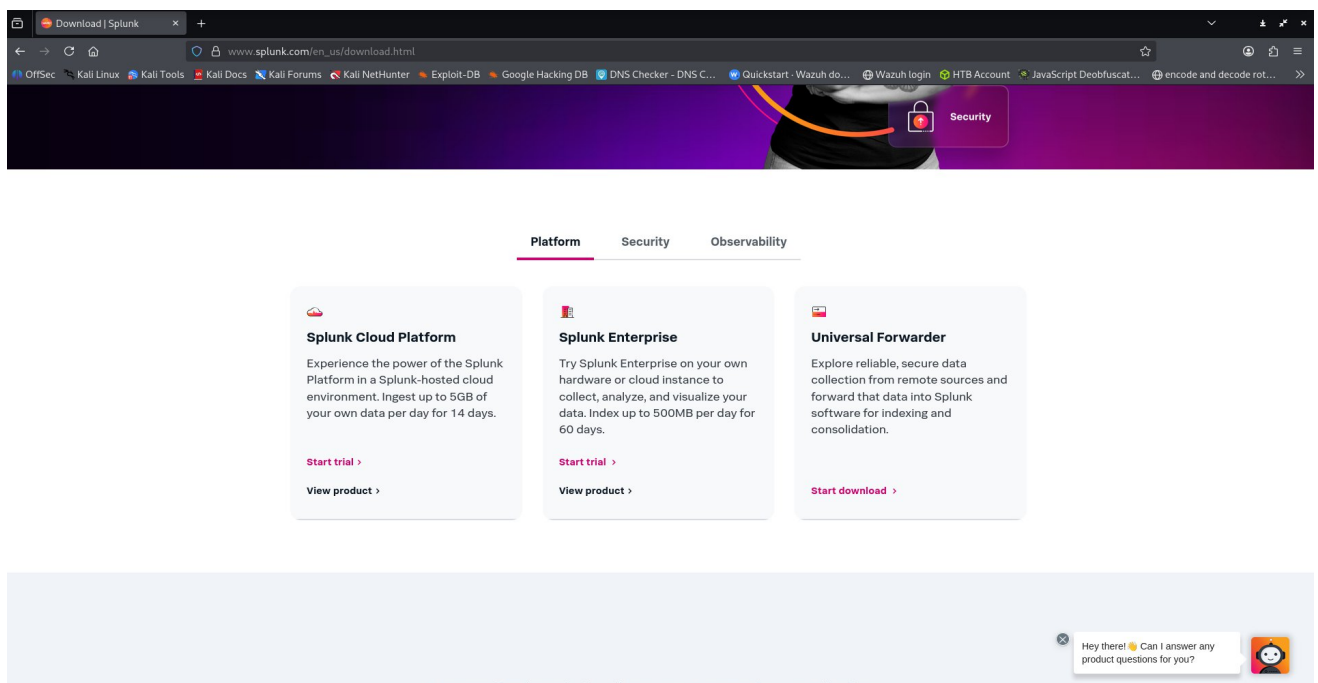
Component	Description
SIEM Tool	Splunk Enterprise (Free 60-days trial)
Operating System	Kali linux
Log Sources Used	auth.log, network.log, firewall.log, malware.log

◆ Setup Steps:

1. Download and install Splunk Enterprise from <https://www.splunk.com>



Click Trials & Downloads options on the webpage, then click splunk enterprise start trial.



We should create and login on the splunk page, after that we could see the splunk enterprise for windows, linux and mac OS. You can choose depends on your Choice. I have selected splunk enterprise for linux

The screenshot shows a web browser window with the Splunk Enterprise download page. The browser's address bar shows the URL: `www.splunk.com/en_us/download/splunk-enterprise.html?locale=en-us`. The page has a navigation bar with links for Platform, Security, Observability, Industries, and Resources. Below the navigation bar is a large orange banner with the text "Choose Your Download". Underneath the banner, the page title is "Splunk Enterprise 10.0.1". A sub-header reads: "Index 500 MB/Day. Sign up and download now. After 60 days you can convert to a perpetual free license or purchase a Splunk Enterprise license to continue using the expanded functionality designed for enterprise-scale deployments." Below this, there is a section titled "Choose Your Installation Package" with tabs for Windows, Linux, and Mac OS. The Linux tab is selected. Under the Linux tab, there is a table with columns for "64-bit", "Operating System", "File Format", "File Size", "Download Now", "Copy wget link", and "More". The table contains one row for "Windows Server 2019, 2022, 2025" with a file size of "915.37 MB". Below the table, there are links for "Release Notes", "System Requirements", "Previous Releases", and "All Other Downloads". At the bottom of the page, there is a footer with four columns: "COMPANY" (About Splunk), "PRODUCTS" (Free Trials & Downloads), "LEARN" (What is SIEM?), and "CONTACT SPLUNK" (Contact Sales).

Splunk Enterprise 10.0.1

Index 500 MB/Day. Sign up and download now. After 60 days you can convert to a perpetual free license or purchase a Splunk Enterprise license to continue using the expanded functionality designed for enterprise-scale deployments.

Choose Your Installation Package

Windows Linux Mac OS

64-bit	Operating System	File Format	File Size	Download Now	Copy wget link	More
	Windows Server 2019, 2022, 2025	.msi	915.37 MB	Download Now	Copy wget link	More

[Release Notes](#) | [System Requirements](#) | [Previous Releases](#) | [All Other Downloads](#)

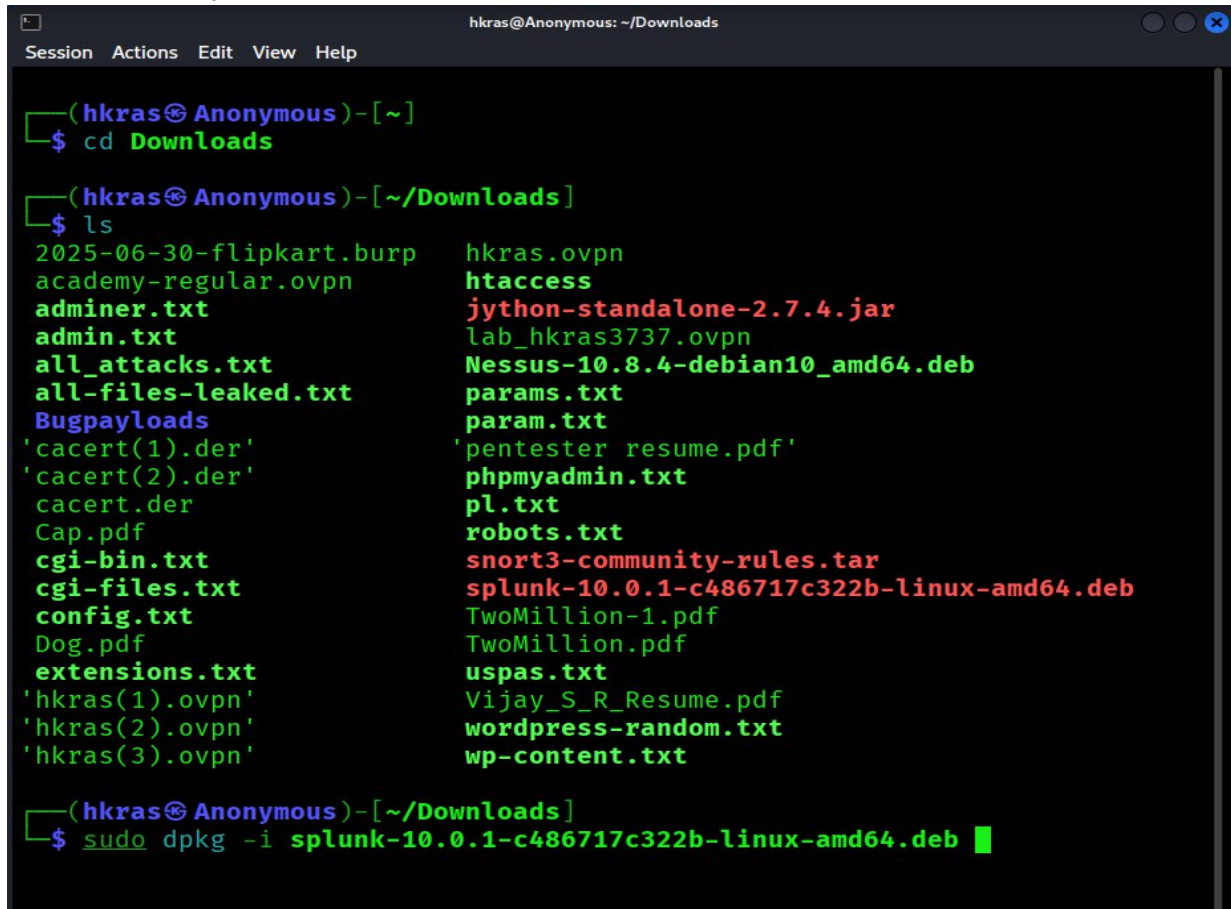
COMPANY
About Splunk

PRODUCTS
Free Trials & Downloads

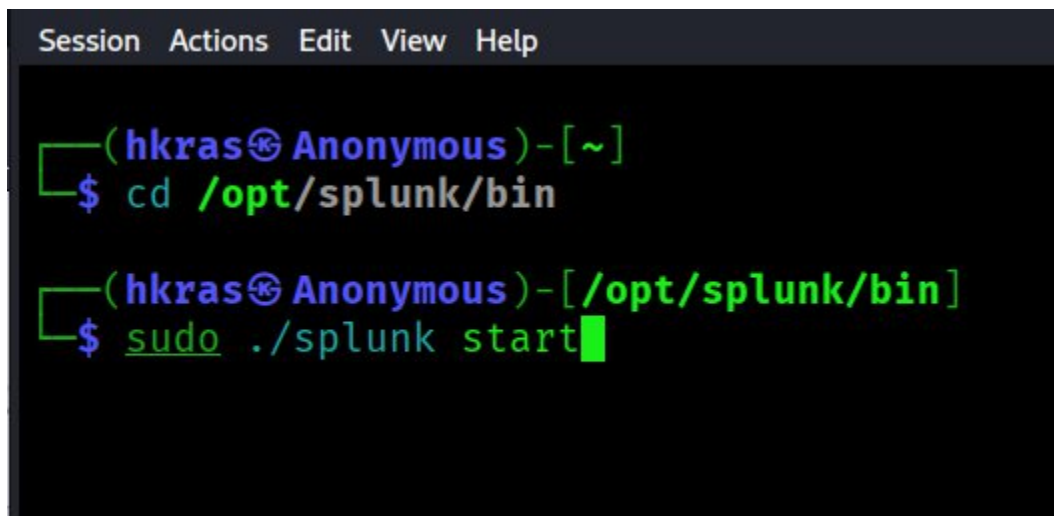
LEARN
What is SIEM?

CONTACT SPLUNK
[Contact Sales](#)

After download finished, we should install splunk enterprise on our system. I installed splunk enterprise on my kali linux vm. Follow the below picture command for splunk installation on kali

A terminal window titled 'hkras@Anonymous: ~/Downloads' with a menu bar (Session, Actions, Edit, View, Help). The prompt is '(hkras@Anonymous)-[~]'. The user enters '\$ cd Downloads'. The prompt changes to '(hkras@Anonymous)-[~/Downloads]'. The user enters '\$ ls', displaying a two-column list of files including various .ovpn, .txt, .pdf, and .deb files. The file 'splunk-10.0.1-c486717c322b-linux-amd64.deb' is highlighted in red. The user then enters '\$ sudo dpkg -i splunk-10.0.1-c486717c322b-linux-amd64.deb'.

Once finished the installation, you should move on to the /opt/splunk/bin directory. And start the splunk using the below picture commands

A terminal window with the same title and menu bar as the previous image. The prompt is '(hkras@Anonymous)-[~]'. The user enters '\$ cd /opt/splunk/bin'. The prompt changes to '(hkras@Anonymous)-[/opt/splunk/bin]'. The user then enters '\$ sudo ./splunk start'.

While running splunk, it will ask agree the license agreement. Put yes and then create administrator username and password, confirm password. Then it will run a webpage like <http://localhost:8000> or <http://127.0.0.1:8000> or otherwise create webpage using user's device name with port number

```
Use Rights: As set out in section 1.1.

Do you agree with this license? [y/n]: n
Do you agree with this license? [y/n]: y

This appears to be your first time running this version of Splunk.

Splunk software must create an administrator account during startup. Otherw
ise, you cannot log in.
Create credentials for the administrator account.
Characters do not appear on the screen when you type in credentials.

Please enter an administrator username: hkras
Password must contain at least:
    * 8 total printable ASCII character(s).
Please enter a new password:
Please confirm new password: 
```

```

hkras@Anonymous: /opt/splunk/bin
Session Actions Edit View Help
.....+.....+.....+...+.....+..+..+...+...+.....+...+..
..+.....+.....+.....+.....+.....+.....+.....+.....+.....+
+..+..+..+.....+.....+.....+.....+.....+.....+.....+.....+
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+++++++
Warning: ignoring -extensions option without -extfile
Certificate request self-signature ok
subject=CN = Anonymous, O = SplunkUser
Done

Waiting for web server at http://127.0.0.1:8000 to be available.....
..... Done

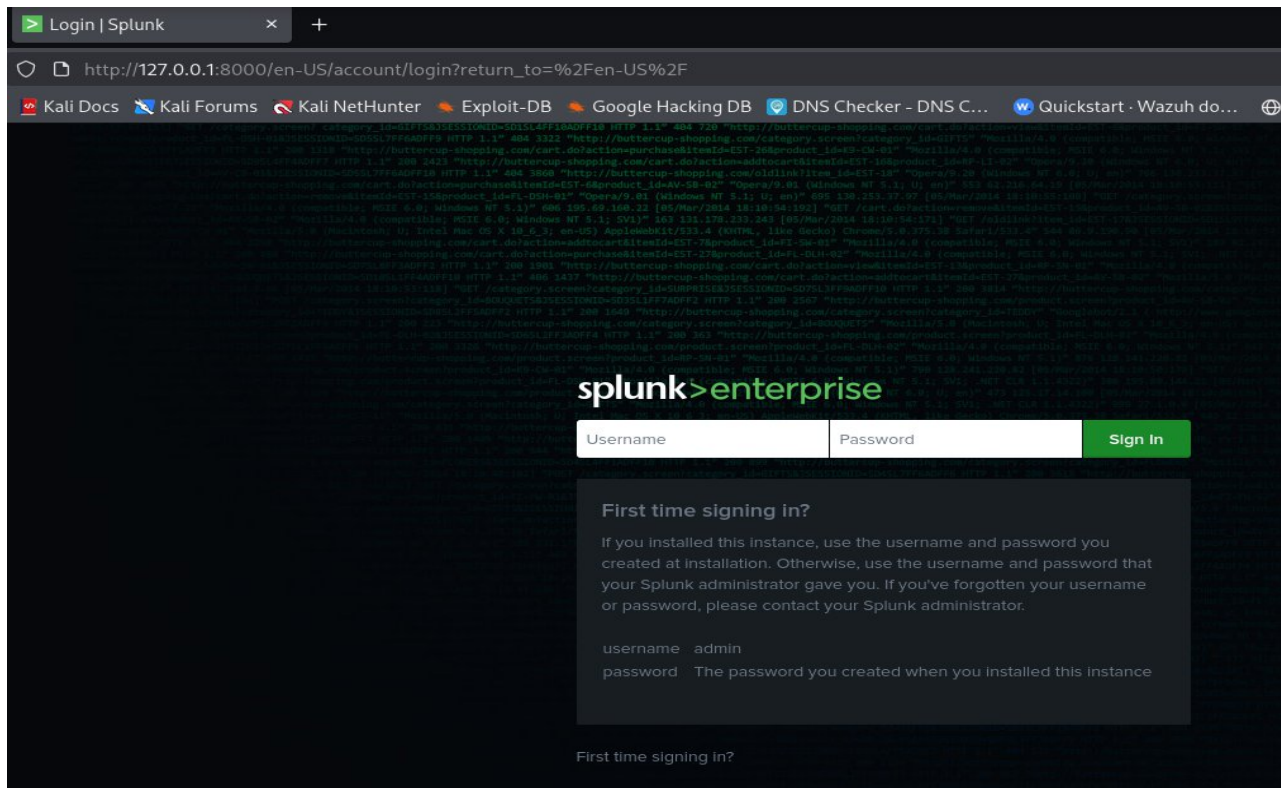
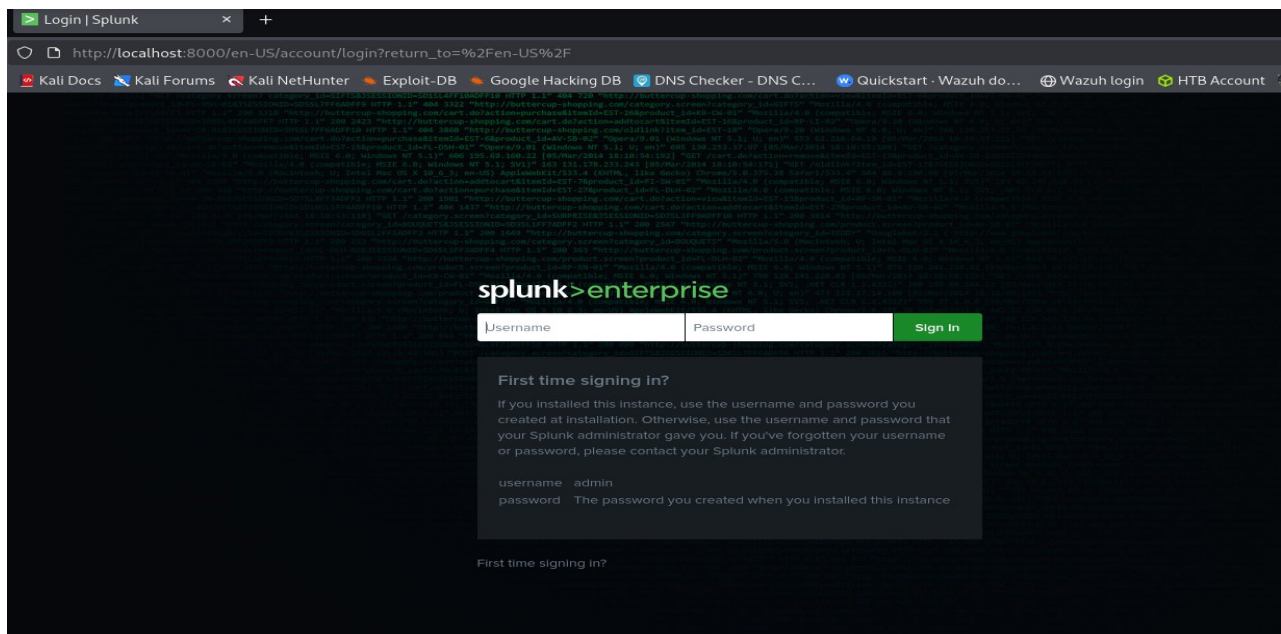
If you get stuck, we're here to help.
Look for answers here: http://docs.splunk.com

The Splunk web interface is at http://Anonymous:8000

(hkras@Anonymous)-[/opt/splunk/bin]
$ █

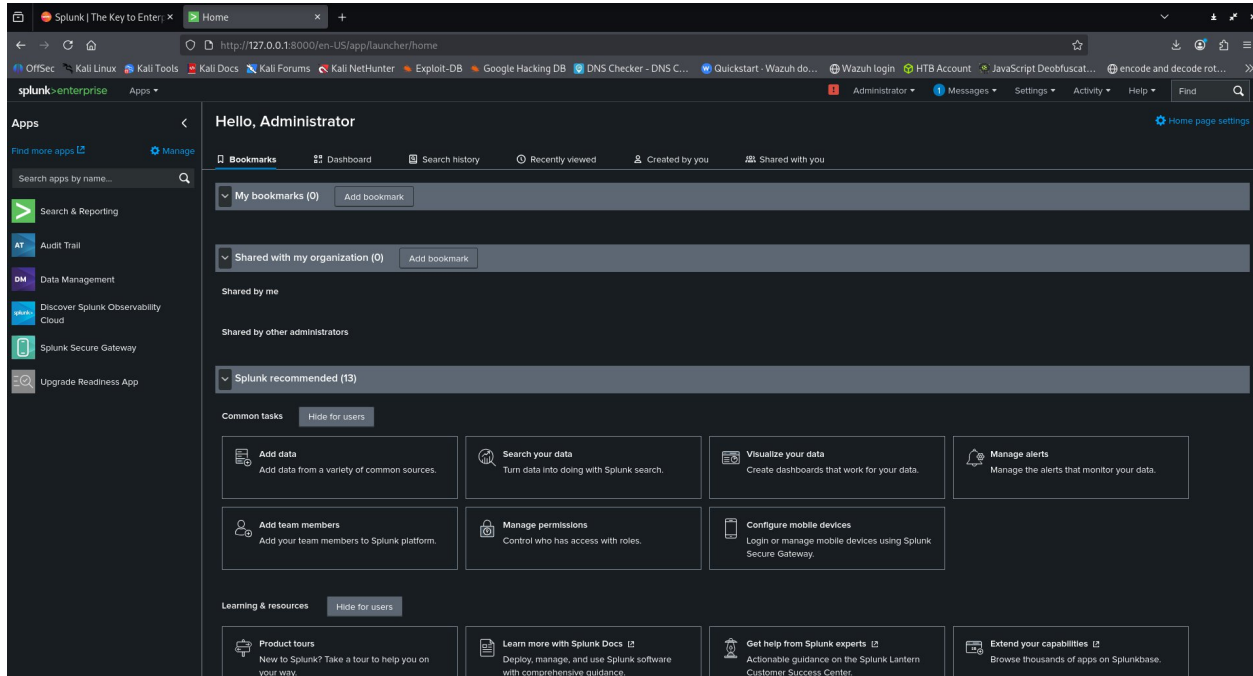
```

Now we should open our browser and type <http://127.0.0.1:8000> or <http://localhost:8000> or <http://our-device-name:8000>

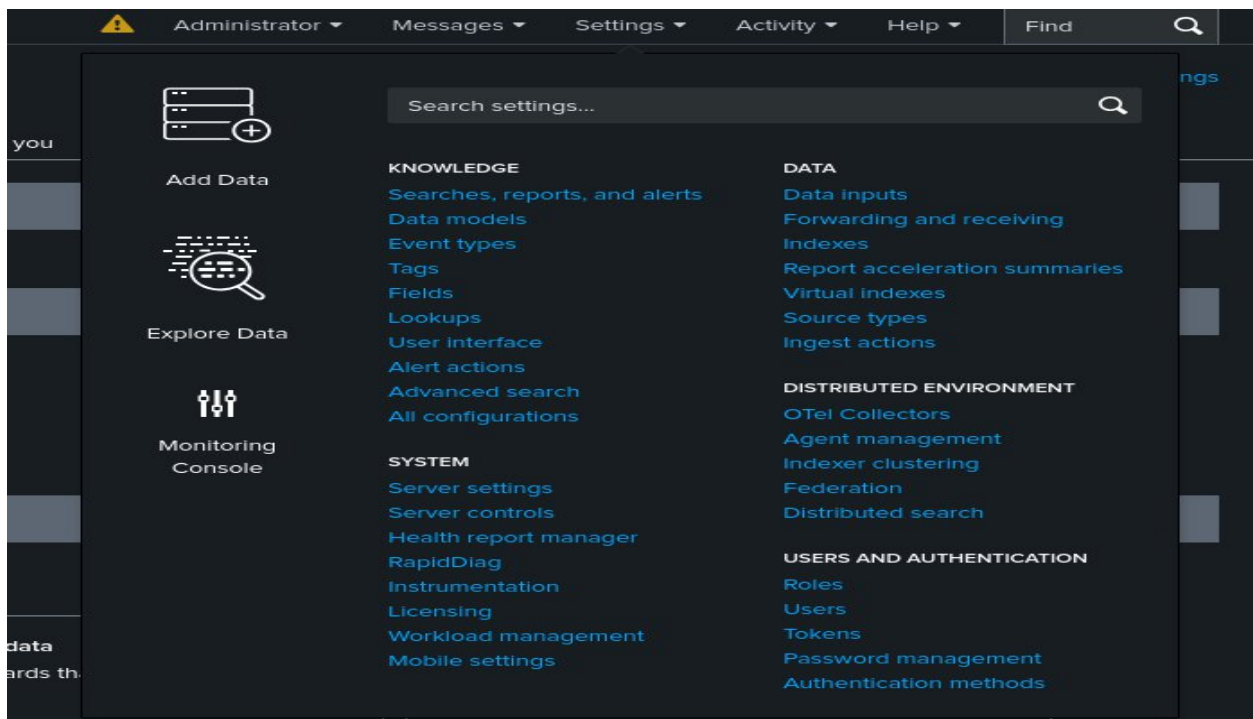


Login with administrator username and password credentials, which you created while starting the splunk.

After successful login, we can see the complete splunk dashboard like below image



We should add necessary .log file into the splunk. Goto settings and click Add Data option



Use the upload option for log file upload

What data do you want to send to the Splunk platform?

Follow guides for onboarding popular data sources

**Cloud computing**
Get your cloud computing data in to the Splunk platform.
10 data sources

**Networking**
Get your networking data in to the Splunk platform.
2 data sources

**Operating System**
Get your operating system data in to the Splunk platform.
1 data source

**Security**
Get your security data in to the Splunk platform.
3 data sources

4 data sources in total

Or get data in with the following methods

**Upload**
files from my computer
Local log files
Local structured files (e.g. CSV)
[Tutorial for adding data](#)

**Monitor**
files and ports on this Splunk platform instance
Files - HTTP - WMI - TCP/UDP - Scripts
Modular inputs for external data sources

**Forward**
data from a Splunk forwarder
Files - TCP/UDP - Scripts

Choose select file option and upload a .log file, and

Add Data - Select Source

← → ↻ ⌂

Not Secure http://anonymous:8000/en-US/manager/search/adddatamethods/selectsource?input_mode=0

OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB DNS Checker - DNS C... Quickstart - Wazuh do... Wazuh login HTB Account

splunk>enterprise Apps Administrator Messages Settings Activity

Add Data

Select Source Set Source Type Input Settings Review Done

< Back Next >

Select Source

Choose a file to upload to the Splunk platform, either by browsing your computer or by dropping a file into the target box below. [Learn More](#)

Selected File: **No file selected**

Select File

Drop your data file here

The maximum file upload size is 500 Mb

FAQ

- > What kinds of files can the Splunk platform index?
- > What is a source?
- > How do I get remote data onto my Splunk platform instance?

Now it will take our .log file name as source

Set Source Type

This page lets you see how the Splunk platform sees your data before indexing. If the events look correct and have the right timestamps, click "Next" to proceed. If not, use the options below to define proper event breaks and timestamps. If you cannot find an appropriate source type for your data, create a new one by clicking "Save As".

Source: **dhcp.log.gz**

Source type: Select Source Type **Save As**

Format	Select...	Select...	Time	Event	1	2	3	4	5	6	7	8	...	Next
1			10/25/25 2:30:00.000 AM	1331981047.238000 2767872470 1331981117.748000 2823389577	CCHNF14C6RA093bP7	192.168.202.76	68	192.168.202.1	67	00:26:9e:83:a2:30	192.168.202.76	0.000000		
2			10/25/25 2:30:00.000 AM	1331981120.628000 7111068 1331981121.800000 4022809768 1331981182.540000 3428947570 1331981216.820000 3106841135 1331981298.710000 288148544 Show all 6 lines	C9svD93TrEvPshf7Gf C2nAD54rx25nILppH CVRJN6491gIrhKzrk C1244g11V2yYf4h1f CXIL6f4ahBkaqMEvc1	192.168.202.102	68	192.168.202.1	67	f0:de:f1:2e:6a:5a	192.168.202.102	0.000000		
3			10/25/25 2:30:00.000 AM	1331981372.700000 2885899026 1331981373.730000	CTeOfd3NNQcegHPmg C1445nrv111GCT7m	192.168.202.101	68	192.168.202.1	67	00:0c:29:8e:38:5a	192.168.202.101	86480.000000		

click next

we should create source type name like this below image

Save Source Type

Name:

Description:

Category:

App:

App should be search & Reporting and save it.

Now you can choose Host field value as your wish and index also.

Add Data - Input Settings

Optional set additional input parameters for this data input as follows:

Host

When the Splunk platform indexes data, each event receives a "host" value. The host value should be the name of the machine from which the event originates. The type of input you choose determines the available configuration options. [Learn More](#)

Host field value:

Index

The Splunk platform stores incoming data as events in the selected index. Consider using a "sandbox" index as a destination if you have problems determining a source type for your data. A sandbox index lets you troubleshoot your configuration without impacting production indexes. You can always change this setting later. [Learn More](#)

Index: [Create a new index](#)

FAQ

- > How do indexes work?
- > How do I know when to create or use multiple indexes?

Click Review and click next click start searching options. Now you can able to see your .log file on real time monitor mode.

New Search

source=\\dhcp.log.gz host=\\Anonymous index=\\main sourcetype=\\dhcp_logs

Time range: All time

333 events (before 10/25/25 2:21:56.000 AM) No Event Sampling

Events (333) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect

1 millisecond per column

Time	Event
10/25/25 2:30:00.000 AM	1332016183.960000 C1Gt9CIV7V2hujMwI 192.168.202.64 68 192.168.202.1 67 c8:60:00:1a:aa:77 192.168.202.64 86400.000000 2952369635
10/25/25 2:30:00.000 AM	1332017407.050000 CLPHSTU4Iw1izzEAS 192.168.202.82 68 192.168.202.1 67 00:0c:29:78:18:23 192.168.202.82 86400.000000 3178925068
10/25/25 2:30:00.000 AM	1332016145.740000 C1XghT1o841EMCRt95 192.168.202.152 68 192.168.202.1 67 c8:60:00:1a:9f:29 192.168.202.152 86400.000000 3965556135
10/25/25 2:30:00.000 AM	1332016154.950000 C1nu8R1ptJxoSuJzq3 192.168.202.97 68 192.168.202.1 67 00:23:54:8a:21:78 192.168.202.97 0.000000 1007804662
10/25/25 2:30:00.000 AM	1332016094.820000 CsSu2i40o7MvnmMue 192.168.202.62 68 192.168.202.1 67 c8:60:00:1a:aa:73 192.168.202.62 86400.000000 2977367047
10/25/25 2:30:00.000 AM	1332015444.490000 CAFZ1E3giXcmPkkad 192.168.202.69 68 192.168.202.1 67 00:0c:20:18:b6:67 192.168.202.69 86400.000000 238122009
10/25/25 2:30:00.000 AM	1332015536.510000 Cn7JDu45yGmM530ke 192.168.202.97 68 192.168.202.1 67 00:23:54:8a:21:78 192.168.202.97 0.000000 2423142510
10/25/25 2:30:00.000 AM	1332015550.620000 CDqQze4Xf8J1v9Qc1 192.168.202.76 68 192.168.202.1 67 00:26:9e:83:a2:30 192.168.202.76 0.000000 2851554795
10/25/25 2:30:00.000 AM	1332015531.030000 C1k42k4M1455555 192.168.202.76 68 192.168.202.1 67 00:26:9e:83:a2:30 192.168.202.76 0.000000 1336585544

You can extract new fields like source ip, destination ip, source port, destination port, user, ip like this.

Select Extract New Fields

The screenshot shows the Splunk Search interface. The top navigation bar includes a search bar with the text "Search | Splunk 10.0.1". Below the navigation bar, there are tabs for "Timeline format", "Zoom Out", and "Zoom to Selection". The main content area is divided into two sections: "SELECTED FIELDS" and "INTERESTING FIELDS".

SELECTED FIELDS:

- `host` 1
- `source` 1
- `sourcetype` 1

INTERESTING FIELDS:

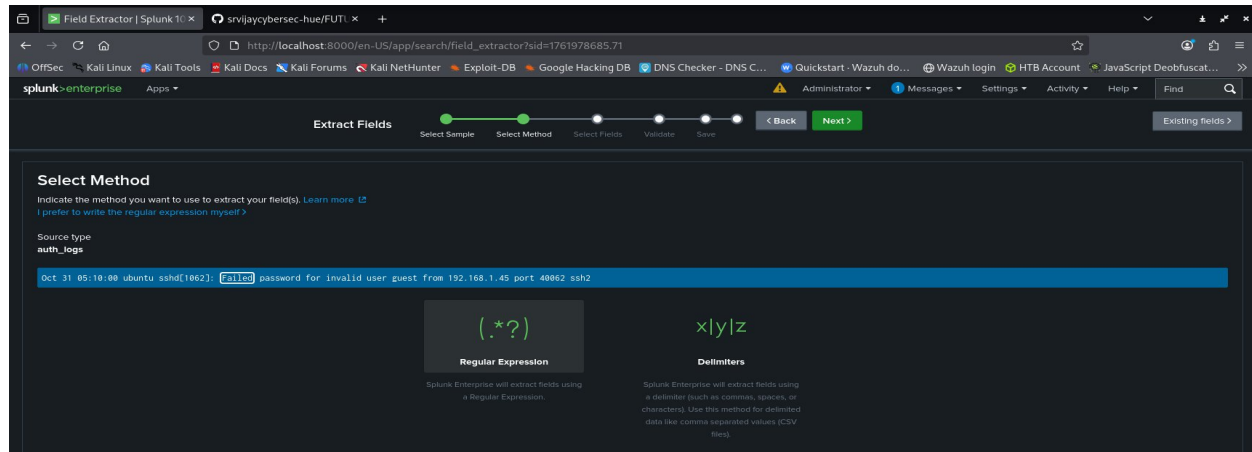
- `date_hour` 1
- `date_mday` 1
- `date_minute` 1
- `date_month` 1
- `date_wday` 1
- `date_year` 1
- `date_zone` 1
- `index` 1
- `linecount` 16
- `punct` 2
- `splunk_server` 1
- `timeendpos` 2
- `timestartpos` 2

At the bottom of the sidebar, there is a button labeled "+ Extract New Fields".

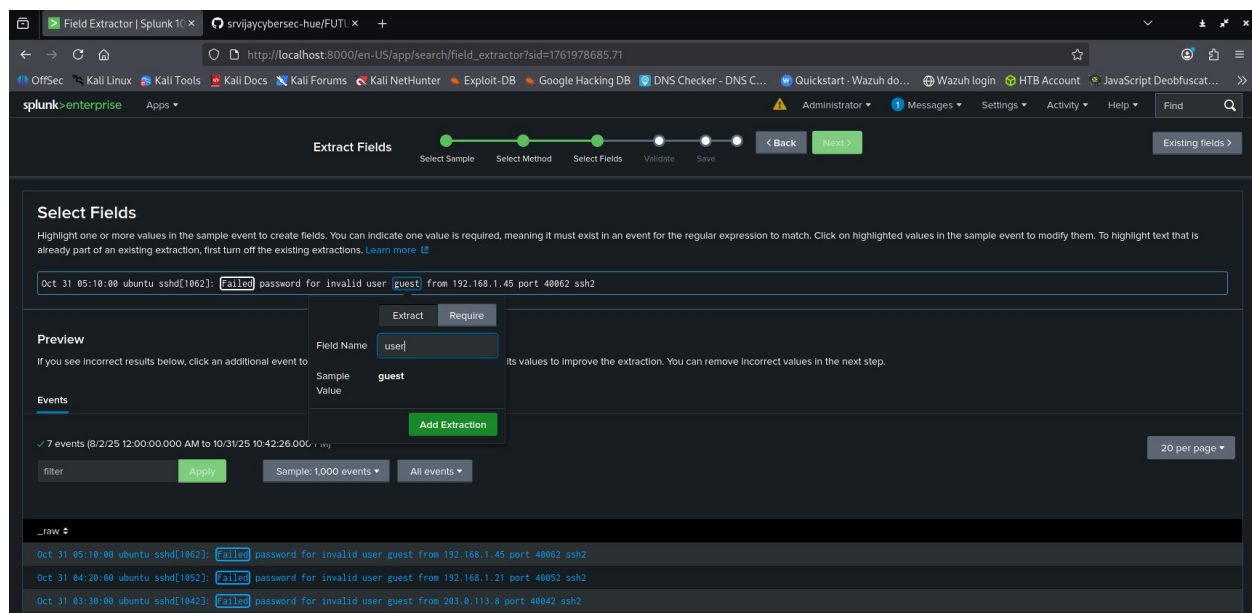
The main table displays search results with the following columns: "Time". The table contains four rows of data, each showing a timestamp: "10/25/25 2:30:00.000 AM".

Time
10/25/25 2:30:00.000 AM
10/25/25 2:30:00.000 AM
10/25/25 2:30:00.000 AM
10/25/25 2:30:00.000 AM

Then select anyone log as a sample click next, choose regular expression and click next



I have choose guest word in the log. It will shows field name, we should give the field name. i gave field name as user



Click next, next, finish....now you can query using your field name with Search Processing Language(SPL)

Now we should upload and create source and source types, extract new fields like this for all .log files.

If you forgot source and source types, go to homepage choose Search & Reporting feature, type on the search bar **| metadata type=sources** we can use **type=sourcetypes**

The screenshot shows the Splunk Enterprise Search & Reporting interface. The search bar contains the query `| metadata type=sources`. The results show 1 result (before 10/30/25 11:59:23.000 PM). The table below displays the search results.

firstTime	lastTime	recentTime	source	totalCount	type
1761388200	1761388200	1761387670	dhcp.log.gz	333	sources

I uploaded auth.log, network.log, firewall.log and malware.log. These are my sources. I have created auth_logs, network_logs, fw_logs and malware_logs as source types on my project.

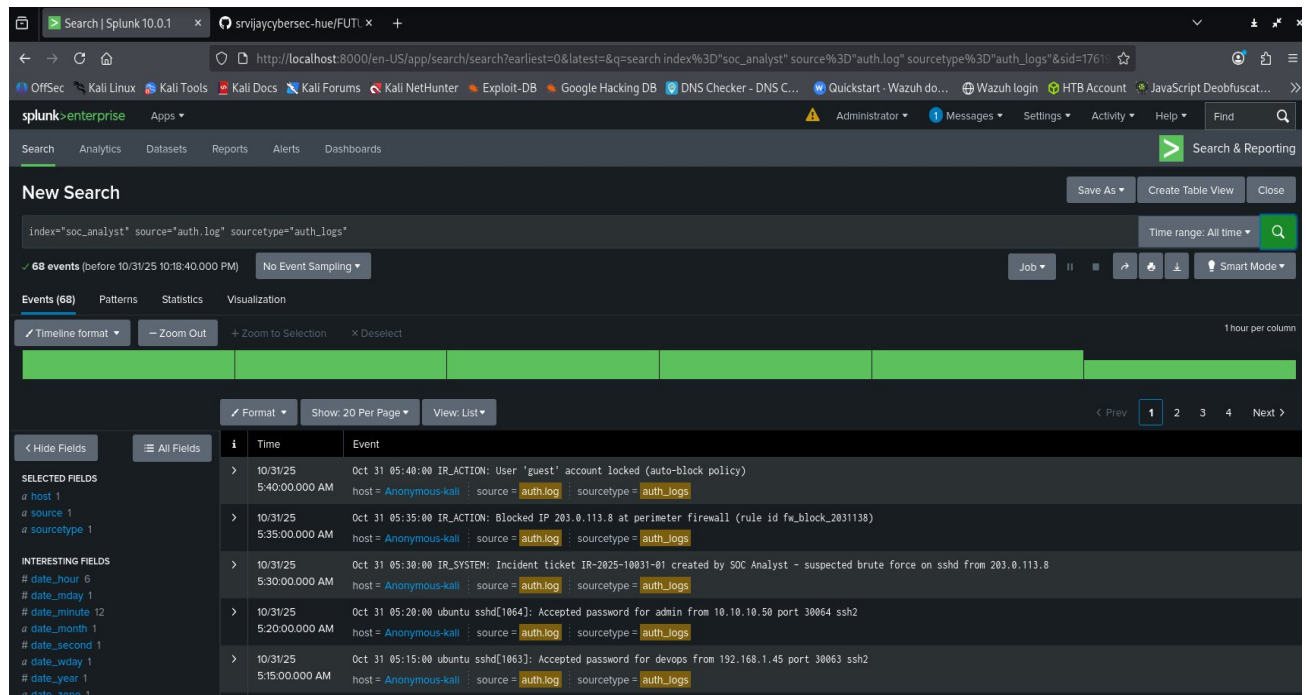
I created index name as soc_analyst while add data

❖ SPL Queries and Analysis

➤ Filter by Source Type

index="soc_analyst" source="auth.log" sourcetype="auth_logs"

Result:



The screenshot shows the Splunk Search interface with the following details:

- Search Bar:** Contains the query `index="soc_analyst" source="auth.log" sourcetype="auth_logs"`.
- Results Summary:** 68 events (before 10/31/25 10:18:40.000 PM).
- Visualization:** Timeline format, zoomed out.
- Fields:** Selected fields include `host`, `source`, and `sourcetype`. Interesting fields include `date_hour`, `date_minute`, `date_month`, `date_second`, `date_wday`, `date_year`, and `date_zone`.
- Event List:** A table of search results.

i	Time	Event
>	Oct 31 05:40:00 5:40:00.000 AM	Oct 31 05:40:00 IR_ACTION: User "guest" account locked (auto-block policy) host = Anonymous-kali : source = auth.log : sourcetype = auth_logs
>	Oct 31 05:35:00 5:35:00.000 AM	Oct 31 05:35:00 IR_ACTION: Blocked IP 203.0.113.8 at perimeter firewall (rule id fw_block_2031138) host = Anonymous-kali : source = auth.log : sourcetype = auth_logs
>	Oct 31 05:30:00 5:30:00.000 AM	Oct 31 05:30:00 IR_SYSTEM: Incident ticket IR-2025-10031-01 created by SOC Analyst - suspected brute force on sshd from 203.0.113.8 host = Anonymous-kali : source = auth.log : sourcetype = auth_logs
>	Oct 31 05:20:00 5:20:00.000 AM	Oct 31 05:20:00 ubuntu sshd[1064]: Accepted password for admin from 10.10.10.50 port 30064 ssh2 host = Anonymous-kali : source = auth.log : sourcetype = auth_logs
>	Oct 31 05:15:00 5:15:00.000 AM	Oct 31 05:15:00 ubuntu sshd[1063]: Accepted password for devops from 192.168.1.45 port 30063 ssh2 host = Anonymous-kali : source = auth.log : sourcetype = auth_logs

➤ Find failed logins

index=_* OR index=* source="auth.log" sourcetype="auth_logs" status=failed

We can use index like this... This is wildcard for search all indexes

Result:

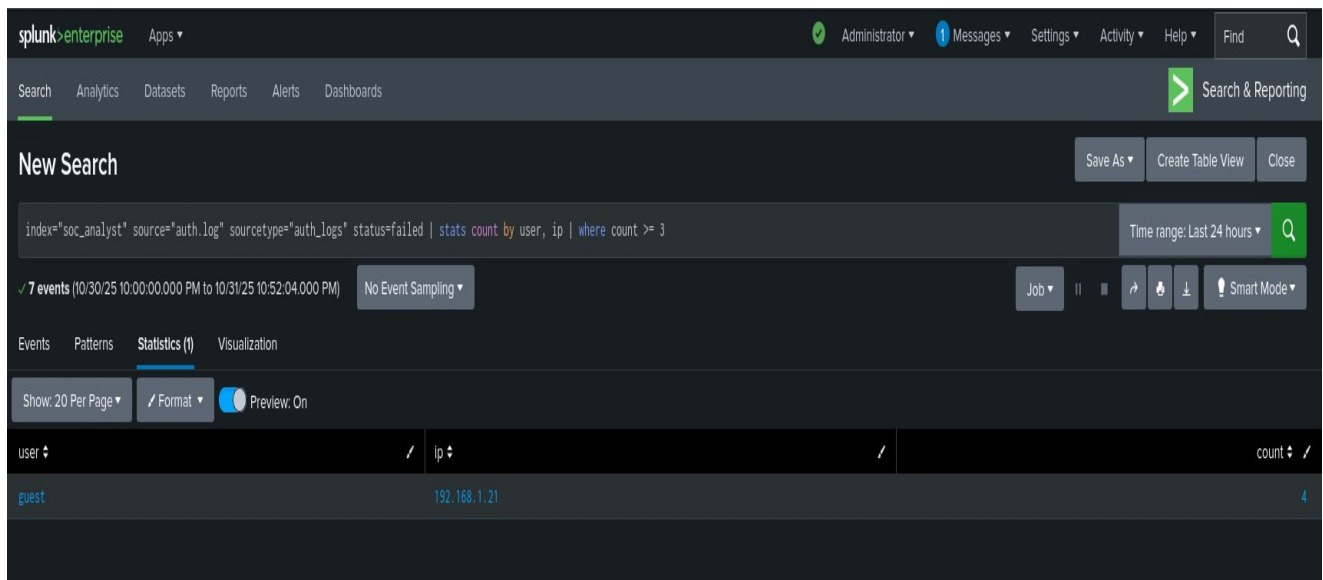
The screenshot shows the Splunk Enterprise interface with a search query: `index=_* OR index=* source="auth.log" sourcetype="auth_logs" status=failed`. The search results show 7 events, all of which are failed login attempts for the 'guest' user from various IP addresses. The events are displayed in a table format, with columns for Time, Event, and Source. The table is filtered to show only events from the last 24 hours.

Time	Event	Source
10/31/25 05:10:00.000 AM	Oct 31 05:10:00 ubuntu sshd[1062]: Failed password for invalid user guest from 192.168.1.45 port 40062 ssh2	host = Anonymous-kali : source = auth.log : sourcetype = auth_logs
10/31/25 04:20:00.000 AM	Oct 31 04:20:00 ubuntu sshd[1052]: Failed password for invalid user guest from 192.168.1.21 port 40052 ssh2	host = Anonymous-kali : source = auth.log : sourcetype = auth_logs
10/31/25 03:30:00.000 AM	Oct 31 03:30:00 ubuntu sshd[1042]: Failed password for invalid user guest from 203.0.113.8 port 40042 ssh2	host = Anonymous-kali : source = auth.log : sourcetype = auth_logs
10/31/25 02:40:00.000 AM	Oct 31 02:40:00 ubuntu sshd[1032]: Failed password for invalid user guest from 192.168.1.21 port 40032 ssh2	host = Anonymous-kali : source = auth.log : sourcetype = auth_logs
10/31/25 01:50:00.000 AM	Oct 31 01:50:00 ubuntu sshd[1022]: Failed password for invalid user guest from 203.0.113.8 port 40022 ssh2	host = Anonymous-kali : source = auth.log : sourcetype = auth_logs

➤ Brute force attempts from same IP

```
index="soc_analyst" source="auth.log" sourcetype="auth_logs" status=failed | stats count by user, ip | where count >= 3
```

Result:



The screenshot shows the Splunk Enterprise interface with a search query: `index="soc_analyst" source="auth.log" sourcetype="auth_logs" status=failed | stats count by user, ip | where count >= 3`. The results are displayed in a table with columns for user, ip, and count. The search found 7 events.

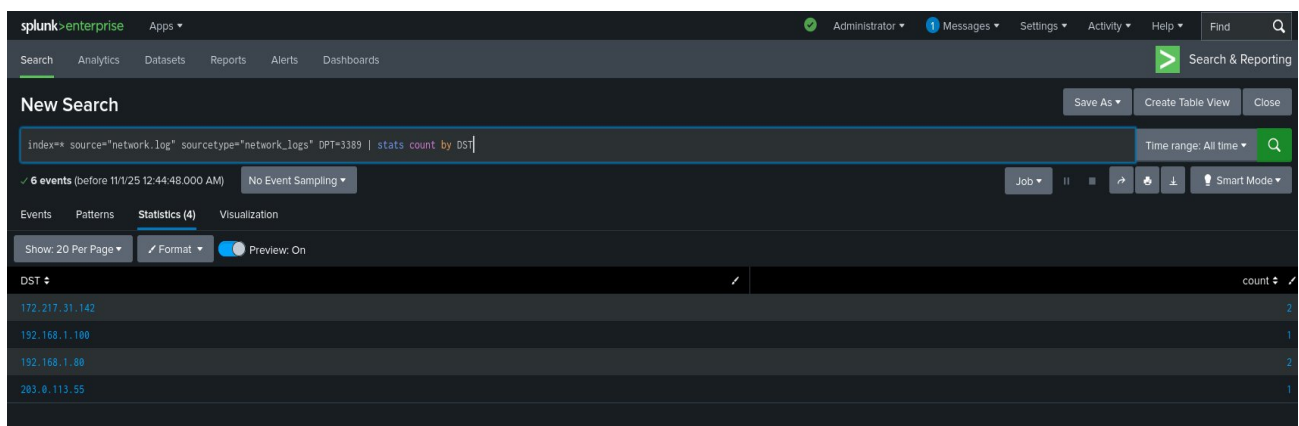
user	ip	count
guest	192.168.1.21	4

➤ Filter for unusual ports

```
index=* source="network.log" sourcetype="network_logs" DPT=3389 | stats count by DST
```

Destination port as DPT, destination as DST

Result:



The screenshot shows the Splunk Enterprise interface with a search query: `index=* source="network.log" sourcetype="network_logs" DPT=3389 | stats count by DST`. The results are displayed in a table with columns for DST and count. The search found 6 events.

DST	count
172.217.31.142	2
192.168.1.100	1
192.168.1.88	2
203.0.113.55	1

➤ High & Critical malware alerts

index="soc_analyst" source="malware.log" sourcetype="malware_logs" severity=High OR severity=Critical

Result:

New Search Save As Create Table View Close

index="soc_analyst" source="malware.log" sourcetype="malware_logs" severity=High OR severity=Critical Time range: All time Q

✓ 13 events (before 11/1/25 12:46:12.000 AM) No Event Sampling Job II ↻ 📄 ⬇ 💡 Smart Mode

Events (13) Patterns Statistics Visualization

✓ Timeline format – Zoom Out + Zoom to Selection X Deselect 1 hour per column

✓ Format Show: 20 Per Page View: List

< Hide Fields ≡ All Fields

SELECTED FIELDS
a host 10
a source 1
a sourcetype 1

INTERESTING FIELDS
a action 6
date_hour 8
date_mday 2
date_minute 9
a date_month 1
date_second 9
a date_wday 2

i	Time	Event
>	10/25/25 12:00:00.000 PM	{ [-] action: isolated event_id: evt-0023 event_type: host_isolation host: corp-win10-02 initiated_by: SOC notes: Host isolated from network after ransomware indicators. sensor_id: sensor-02 severity: critical threat_name: isolation_for_remediation timestamp: 2025-10-26T01:30:00+05:30 } Show as raw text

:

➤ Blocked connections by source IP

index="soc_analyst" source="firewall.log" sourcetype="fw_logs" UFW BLOCK

Result:

The screenshot shows the Splunk Enterprise interface with a search query: `index="soc_analyst" source="firewall.log" sourcetype="fw_logs" UFW BLOCK`. The search results show 14 events. The timeline view shows a series of green bars representing blocked connections. The table view shows the following events:

Time	Event
10/31/25 4:20:00.000 AM	Oct 31 04:20:00 Firewall101 kernel: [UFW BLOCK] IN=eth1 OUT= MAC=00:50:56:c0:00:41 SRC=198.51.100.50 DST=192.168.1.50 LEN=52 PROTO=TCP SPT=40065 DPT=22 WINDOW=65535 SYN host = Anonymous-kali : source = firewall.log : sourcetype = fw_logs
10/31/25 4:08:00.000 AM	Oct 31 04:08:00 Firewall101 kernel: [UFW BLOCK] IN=eth1 OUT= MAC=00:50:56:c0:00:3e SRC=198.51.100.50 DST=192.168.1.100 LEN=60 PROTO=TCP SPT=40062 DPT=443 WINDOW=65535 SYN host = Anonymous-kali : source = firewall.log : sourcetype = fw_logs
10/31/25 3:48:00.000 AM	Oct 31 03:48:00 Firewall101 kernel: [UFW BLOCK] IN=eth1 OUT= MAC=00:50:56:c0:00:39 SRC=203.0.113.9 DST=192.168.1.1 LEN=60 PROTO=TCP SPT=40057 DPT=161 WINDOW=65535 SYN host = Anonymous-kali : source = firewall.log : sourcetype = fw_logs
10/31/25 3:36:00.000 AM	Oct 31 03:36:00 Firewall101 kernel: [UFW BLOCK] IN=eth1 OUT= MAC=00:50:56:c0:00:36 SRC=198.51.100.50 DST=192.168.1.50 LEN=52 PROTO=TCP SPT=40054 DPT=1234 WINDOW=65535 SYN host = Anonymous-kali : source = firewall.log : sourcetype = fw_logs
10/31/25 3:24:00.000 AM	Oct 31 03:24:00 Firewall101 kernel: [UFW BLOCK] IN=eth1 OUT= MAC=00:50:56:c0:00:33 SRC=192.168.1.45 DST=192.168.1.100 LEN=52 PROTO=TCP SPT=40051 DPT=445 WINDOW=65535 SYN host = Anonymous-kali : source = firewall.log : sourcetype = fw_logs
10/31/25	Oct 31 03:20:00 Firewall101 kernel: [UFW BLOCK] IN=eth1 OUT= MAC=00:50:56:c0:00:32 SRC=198.51.100.50 DST=192.168.1.100 LEN=52 PROTO=TCP SPT=40050 DPT=22 WINDOW=65535 SYN

➤ External authentication attempts

index="soc_analyst" source="auth.log" sourcetype="auth_logs" | search NOT ip="192.168.*"

Result:

The screenshot shows the Splunk Enterprise interface with a search query: `index="soc_analyst" source="auth.log" sourcetype="auth_logs" | search NOT ip="192.168.*"`. The search results show 63 events. The timeline view shows a series of green bars representing authentication attempts. The table view shows the following events:

Time	Event
10/31/25 5:40:00.000 AM	Oct 31 05:40:00 IR_ACTION: User 'guest' account locked (auto-block policy) host = Anonymous-kali : source = auth.log : sourcetype = auth_logs
10/31/25 5:35:00.000 AM	Oct 31 05:35:00 IR_ACTION: Blocked IP 203.0.113.8 at perimeter firewall (rule id fw_block_2031138) host = Anonymous-kali : source = auth.log : sourcetype = auth_logs
10/31/25 5:30:00.000 AM	Oct 31 05:30:00 IR_SYSTEM: Incident ticket IR-2025-10031-01 created by SOC Analyst - suspected brute force on sshd from 203.0.113.8 host = Anonymous-kali : source = auth.log : sourcetype = auth_logs
10/31/25 5:20:00.000 AM	Oct 31 05:20:00 ubuntu sshd[1064]: Accepted password for admin from 10.10.10.50 port 30064 ssh2 host = Anonymous-kali : source = auth.log : sourcetype = auth_logs
10/31/25 5:15:00.000 AM	Oct 31 05:15:00 ubuntu sshd[1063]: Accepted password for devops from 192.168.1.45 port 30063 ssh2 host = Anonymous-kali : source = auth.log : sourcetype = auth_logs

➤ Frequent firewall blocks

index="soc_analyst" source="firewall.log" sourcetype="fw_logs" UFW BLOCK | stats count by SRC | sort - count

Result:

The screenshot shows the Splunk Enterprise interface with a search bar containing the query: `index="soc_analyst" source="firewall.log" sourcetype="fw_logs" UFW BLOCK | stats count by SRC | sort - count`. The search results are displayed in a table with 5 rows, showing the source IP (SRC) and the count of events.

SRC	count
198.51.100.50	4
192.168.1.45	3
192.168.1.77	3
203.0.113.9	3
198.51.100.23	1

➤ Failed logins by user

index=soc_analyst source="auth.log" sourcetype="auth_logs" status=failed | stats count by user

Result:

The screenshot shows the Splunk Enterprise interface with a search bar containing the query: `index=soc_analyst source="auth.log" sourcetype="auth_logs" status=failed | stats count by user`. The search results are displayed in a table with 1 row, showing the user and the count of events.

user	count
guest	7

➤ Malware remediation

index="soc_analyst" source="malware.log" sourcetype="malware_logs"
action=quarantined

Result:

The screenshot shows the Splunk Enterprise interface with a search for `index="soc_analyst" source="malware.log" sourcetype="malware_logs" action=quarantined`. The search results show 3 events. The first event is expanded, showing details for a scheduled task execution that was quarantined.

Time	Event
10/25/25 5:30:00.000 AM	<pre>{ [-] action: quarantined event_id: evt-0017 event_type: suspicious_scheduled_task_run host: corp-win10-10 notes: Scheduled task executed unknown updater binary; quarantined. pid: 3456 process_name: Updater.exe sensor_id: sensor-13 severity: medium task_name: Microsoft\Windows\Updater threat_name: persistence_execution timestamp: 2025-10-25T19:00:00+05:30 user: CORPUser }</pre>

➤ Unusual network ports

index="soc_analyst" source="network.log" sourcetype="network_logs" NOT DPT=22
NOT DPT=80 NOT DPT=443

Result:

The screenshot shows the Splunk Enterprise interface with a search for `index="soc_analyst" source="network.log" sourcetype="network_logs" NOT DPT=22 NOT DPT=80 NOT DPT=443`. The search results show 55 events. The first event is expanded, showing details for a network action that isolated a host on a specific port.

Time	Event
10/31/25 3:50:00.000 AM	<pre>Oct 31 03:50:00 IR_ACTION: Isolated host 192.168.1.20 on switch port (quarantine) - ticket IR-2025-10031-02 host = Anonymous-kali source = network_log sourcetype = network_logs</pre>
10/31/25 3:45:00.000 AM	<pre>Oct 31 03:45:00 IR_ACTION: Network ACL applied to block outbound to 172.217.31.142 on port 8080 for host 192.168.1.20 host = Anonymous-kali source = network_log sourcetype = network_logs</pre>
10/31/25 3:42:00.000 AM	<pre>Oct 31 03:42:00 IDS01 kernel: IN=eth0 OUT= MAC=00:0c:29:4d:8e:4a SRC=10.0.0.5 DST=8.8.8.8 LEN=60 PROTO=TCP SPT=40074 DPT=445 SYN host = Anonymous-kali source = network_log sourcetype = network_logs</pre>
10/31/25 3:40:00.000 AM	<pre>Oct 31 03:40:00 IR_SYSTEM: SOC Analyst created IOC list and added 203.0.113.8, cmd-control.evil.com to threat feed host = Anonymous-kali source = network_log sourcetype = network_logs</pre>
10/31/25 3:39:00.000 AM	<pre>Oct 31 03:39:00 IDS01 kernel: IN=eth0 OUT= MAC=00:0c:29:4d:8e:4a SRC=10.0.0.22 DST=192.168.1.80 LEN=60 PROTO=TCP SPT=40073 DPT=3389 SYN host = Anonymous-kali source = network_log sourcetype = network_logs</pre>
10/31/25 3:36:00.000 AM	<pre>Oct 31 03:36:00 IDS01 snort[472]: [1:100072:1] 50_ Injection attempt detected [Priority: 1] (TCP) 10.0.0.45:55411 -> 192.168.1.100:80</pre>

❖ Incident Response Simulation

Incident type	Description	Response Action
Brute force attack (or) login failure	Repeated SSH login failures from same IP address	Block IP in firewall; reset user passwords
Port Scanning	Unusual port access patterns detected	Identify attacker IP and block network segment
Malware Infection	Suspicious executable detected	Isolate host; run antivirus and quarantine or delete malicious files

✓ Conclusion:

Splunk effectively helped identify and simulate SOC activities such as log monitoring, threat detection, and incident response.

This task successfully demonstrates SOC workflow for beginner analysts.

❖ Future Improvements

- Integrate Splunk SOAR for automated response actions.
- Include threat intelligence feeds for IP reputation checking.
- Connect real-time syslog sources for continuous monitoring

❖ Final Note

- Security event correlation
- Threat detection logic using SPL
- Incident Management lifecycle

Result: Successfully completed “SOC Alert Monitoring & Incident Response Simulation” using Splunk Enterprise.